

LLM-enhanced Federated Graph Learning with Geometry-aware Graph Projection and Shared Subspace Aggregation

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Abstract

Federated graph learning (FGL) enables collaborative training without sharing raw graph data, but existing methods remain limited when clients perform different types of graph learning tasks. Recent advances in large language models (LLMs) provide new opportunities, where a promising solution is to project graph data into the LLM input space and leverage their generalization ability for prediction. However, directly applying LLMs to FGL raises two key challenges: (1) enhancing LLM interpretation of graphs with limited client data, and (2) obtaining a generalizable graph projector across heterogeneous client tasks. To address these challenges, we propose a LLM-enhanced federated graph learning framework FedLGS, which consists of two modules, i.e., geometry-aware graph projection (GGP) and shared subspace aggregation (SSA). GGP projects graph representations into complementary Riemannian manifolds with Ollivier-Ricci curvature priors, enabling geometry-aware and semantically enriched inputs for LLMs. SSA applies curvature-weighted whitening and aggregates updates in a shared subspace to achieve geometry-consistent generalization across heterogeneous tasks. Extensive experiments on six datasets demonstrate the effectiveness of FedLGS.

CCS Concepts

• Information systems → Data mining; • Computing methodologies → Distributed artificial intelligence; Natural language processing.

Keywords

Graph Learning, Federated Learning, Large Language Models

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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792546>

ACM Reference Format:

Pengyang Zhou, Zhihao Huang, Jiahe Xu, Wu Wen, Xiaolin Zheng, Chaochao Chen, and Jianwei Yin. 2026. LLM-enhanced Federated Graph Learning with Geometry-aware Graph Projection and Shared Subspace Aggregation. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792546>

1 Introduction

Graph-structured data is ubiquitous in various real-world applications such as e-commerce [56] and social networks [8]. In the real world, different organizations usually maintain their own graph data, while privacy regulations [52] prohibit direct data sharing. These constraints have motivated federated graph learning (FGL) [12, 32], which enables collaborative model training while keeping data local. Recent research has increasingly focused on addressing the non-independent and identically distributed (non-IID) issue in FGL [2, 18, 25, 61], intending to ensure robust model performance under heterogeneous client data distributions.

However, existing FGL methods fail to generalize across task-heterogeneous clients. As illustrated in Figure 1(a), current approaches typically rely on jointly training graph neural networks (GNNs), which assume the same learning objective across clients. This assumption becomes problematic when heterogeneous tasks like node classification and link prediction coexist. For example, in social networks, one client may focus on predicting user interest categories, while another aims to infer whether two users should be connected as friends. Such limitations motivate the development of a more flexible framework that can adapt to client diversity.

Recently, large language models (LLMs) have shown strong generalization ability for graph learning [20, 57]. Pretrained on massive corpora, LLMs provide a rich semantic space that can be adapted to a wide range of tasks [23]. Current studies can be grouped into two major directions. *First*, LLMs are incorporated as a powerful enhancer, where they inject semantic priors into node representations [31, 67] or provide reasoning and task-aware supervision [9, 15]. Although these approaches enrich embeddings and

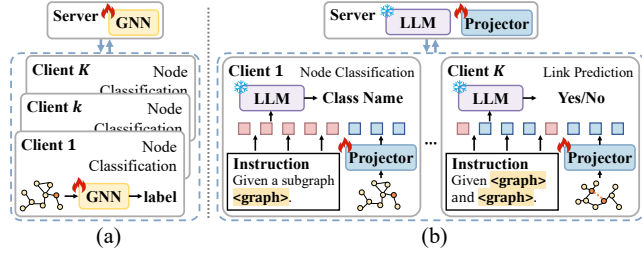


Figure 1: (a) Existing FGL methods often assume uniform GNN objectives across clients. (b) Leveraging LLMs in FGL enables the handling of task-heterogeneous clients.

improve interpretability, their downstream GNN predictors remain task-specific. Differences in supervision granularity and output space make it challenging to accommodate multiple heterogeneous tasks within one model [49]. *Second*, recent work explores directly leveraging the predictive capability of LLMs [6, 50]. As illustrated in Figure 1(b), node information and graph structure, together with task-specific instructions, are tokenized and fed into an LLM to generate predictions. This design allows LLMs to process graph-structured information in a unified language space. Its extension to federated training is particularly promising for tackling diverse graph-related tasks across multiple clients.

Nevertheless, directly leveraging LLMs in FGL is non-trivial and challenging. Firstly, *How to enhance LLM interpretation of graphs with limited client data? (CH1)* In FGL settings, each client typically holds only a small set of labeled samples. This contrasts with centralized training, where LLMs can leverage abundant and diverse annotations. The limited client local data severely restricts the ability of LLMs to learn meaningful graph representations. Moreover, existing approaches often assume that graph data can be fully represented in a single Euclidean space [6, 50], which oversimplifies the underlying geometry [46] and prevents LLMs from achieving a complete view of graphs. Therefore, it is necessary to construct more informative graph representations that enable LLMs to interpret graphs under limited supervision.

Secondly, *How to obtain a generalizable graph projector across heterogeneous task objectives? (CH2)* To enable LLM-based prediction, each client trains a projector that maps local graph representations into the LLM input space. Since these projectors are optimized under distinct task objectives and supervision signals, they inevitably encode heterogeneous inductive biases and emphasize different semantic aspects of the data. This discrepancy will also appear across each parameter dimension, reflecting uneven scaling under different client local geometric conditions. Consequently, conventional parameter averaging [37] struggles to integrate these heterogeneous client updates into a geometry-consistent global graph projector model. This motivates the need for an aggregation strategy that constructs a consistent shared subspace, capable of supporting generalization across heterogeneous objectives.

To address these challenges, we propose a novel LLM-enhanced federated graph learning framework named FedLGS, which consists of two modules, i.e., **geometry-aware graph projection (GGP)** and **shared subspace aggregation (SSA)**. To address CH1, GGP projects node representations into three complementary Riemannian manifolds, each designed to capture distinct geometric properties. Each client leverages its precomputed Ollivier-Ricci curvature to refine

node representations within these manifolds. These geometrical representations are then mapped into the LLM’s input space as semantically enriched tokens, improving the ability of LLM to interpret graph-structured data under limited supervision. To address CH2, SSA mitigates heterogeneity by first collecting curvature-based importance vectors from clients to construct geometry-aware whitening metrics. These metrics calibrate parameter sensitivity of the graph projector and enable the extraction of a consistent shared subspace via Singular Value Decomposition (SVD). Client updates are then projected onto this subspace, and their aligned components are aggregated to form a global graph projector, which effectively minimizes interference from heterogeneous local objectives while preserving geometric consistency across the federated system.

The main contributions are summarized as follows: (1) We are the first to introduce LLM-enhanced federated graph learning that effectively handles task heterogeneity across clients, enabling generalized graph understanding under diverse learning objectives. (2) We propose a framework FedLGS consisting of GGP and SSA. GGP projects graph representations into complementary Riemannian manifolds with edge curvature priors, enabling geometry-aware and semantically enriched inputs for LLMs. Meanwhile, SSA applies curvature-weighted whitening and aggregates updates in a shared subspace, achieving geometry-consistent generalization across heterogeneous tasks. (3) We conduct extensive experiments on six datasets and validate the effectiveness of FedLGS.

2 Related Work

2.1 Federated Graph Learning

Federated learning enables multiple clients to collaboratively train models without sharing raw data [4, 28, 33, 37]. Federated graph learning (FGL) extends this paradigm to graph-structured domains, where information is inherently relational [12, 32, 60]. Based on prediction targets, existing work can be divided into graph-level, node-level, and subgraph-level tasks. Graph-level methods mainly address non-IID label distributions [47, 48, 58], fairness [42], and OOD generalization [10, 29, 63], paralleling challenges in conventional federated learning. Node-level tasks typically assume each client is a node with access to its own features and neighbors, a setting common in recommendation systems where users act as clients with interaction histories [35, 36, 45, 55, 62, 65]. In this work, we focus on the subgraph-level task, where each client can perform node classification or link prediction. Many existing methods are tailored to node classification, which relies on node label information [11, 18, 25, 66]. In contrast, another line of work modifies the aggregation process, measuring functional similarity and applying personalized parameter masks [2], introducing curriculum-guided updates with structural indicators [22], leveraging cross-client knowledge distillation [43, 64], or incorporating topology-aware weighting factors [27]. However, these methods remain limited in handling task heterogeneity, where clients not only differ in data distributions but also pursue different learning objectives.

2.2 Large Language Models for Graphs

Large language models (LLMs) have recently attracted growing attention in graph learning due to their strong generalization and

reasoning abilities. *First*, a number of studies integrate LLMs with GNNs, either by injecting semantic knowledge into node representations [13, 16, 31, 67] or leveraging LLM-based reasoning and planning to generate explanations or task-specific supervisory signals [7, 9, 15]. These methods enrich embeddings and enhance discriminative power, yet they still rely on GNNs as the core predictor, which limits their adaptability to heterogeneous tasks. *Second*, more recent work attempts to exploit LLMs directly for prediction, where textual and structural features of nodes are transformed into token sequences, combined with task instructions, and subsequently processed by an LLM to produce outputs [5, 6, 16, 26, 50, 53]. Such a design enables LLMs to interpret graph data and generate predictions, effectively unifying structural and semantic information to support diverse graph tasks. However, its extension to federated settings remains unexplored, where limited client data restricts effective graph interpretation and heterogeneous task objectives make it difficult to obtain a generalizable projector across clients.

3 Preliminaries

3.1 Riemannian Manifold

A Riemannian manifold $(\mathcal{M}, g)$ is a smooth manifold $\mathcal{M}$ endowed with a Riemannian metric g [30]. At each point $\mathbf{x} \in \mathcal{M}$, the tangent space $\mathcal{T}_{\mathbf{x}}\mathcal{M}$ is a Euclidean vector space, and the local metric $g_{\mathbf{x}}(\cdot, \cdot) : \mathcal{T}_{\mathbf{x}}\mathcal{M} \times \mathcal{T}_{\mathbf{x}}\mathcal{M} \rightarrow \mathbb{R}$ defines inner products, thereby inducing notions of length, distance, and angle. The exponential map $\exp_{\mathbf{x}} : \mathcal{T}_{\mathbf{x}}\mathcal{M} \rightarrow \mathcal{M}$ projects tangent vectors along geodesics onto the manifold, while the logarithmic map $\log_{\mathbf{x}} : \mathcal{M} \rightarrow \mathcal{T}_{\mathbf{x}}\mathcal{M}$ provides the inverse transformation back to the tangent space.

The κ -stereographic model [1] offers a unified representation of constant-curvature manifolds:

$$\mathcal{M}^{\kappa, d} = \{\mathbf{x} \in \mathbb{R}^d \mid -\kappa \|\mathbf{x}\|_2^2 < 1\}, \quad (1)$$

where d is the intrinsic dimension of the manifold. The Riemannian metric in this model is conformal to the Euclidean metric:

$$g_{\mathbf{x}}(\mathbf{u}, \mathbf{v}) = (\lambda_{\mathbf{x}}^{\kappa})^2 \langle \mathbf{u}, \mathbf{v} \rangle, \quad \lambda_{\mathbf{x}}^{\kappa} = \frac{2}{1 + \kappa \|\mathbf{x}\|_2^2}. \quad (2)$$

Here, $\lambda_{\mathbf{x}}^{\kappa}$ is the conformal factor that rescales the Euclidean inner product according to curvature κ . This framework subsumes hyperbolic space $(\mathbb{H}^d, \kappa < 0)$, spherical space $(\mathbb{S}^d, \kappa > 0)$, and Euclidean space $(\mathbb{E}^d, \kappa = 0)$. Further details on the fundamental operations are provided in Appendix E.

3.2 Ollivier-Ricci Curvature

In graphs, the absence of a smooth manifold structure requires discrete analogs of curvature. Ollivier-Ricci curvature [41] extends the classical notion of Ricci curvature [51] to networks by assigning curvature values to edges, defined through the transportation distance between neighborhood measures. For an unweighted graph, the neighborhood distribution of a node i is given by a lazy random walk:

$$\xi_i^{\delta}(z) = \begin{cases} \delta, & z = i, \\ \frac{1-\delta}{|\mathcal{N}(i)|}, & z \in \mathcal{N}(i), \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

where $\delta \in [0, 1]$ controls the probability of staying at node i , and $\mathcal{N}(i)$ denotes the set of one-hop neighbors of node i . The Ollivier-Ricci curvature of an edge (i, j) is then defined as

$$\rho(i, j) = 1 - \frac{W_1(\xi_i^{\delta}, \xi_j^{\delta})}{d_{\mathcal{G}}(i, j)}, \quad (4)$$

where W_1 denotes the Wasserstein-1 distance [44] and $d_{\mathcal{G}}(i, j)$ is the graph distance, i.e., the length of the shortest path between i and j . Further details are provided in Appendix F.

4 Method

4.1 Problem Statement

We consider a federated graph learning system with a set of clients $\{1, 2, \dots, K\}$. Each client k owns a local graph $\mathcal{G}_k = (\mathcal{V}_k, \mathcal{E}_k, \mathbf{X}_k)$, where $\mathcal{V}_k$ is the node set, $\mathcal{E}_k \subseteq \mathcal{V}_k \times \mathcal{V}_k$ is the edge set encoding the connectivity among nodes, and $\mathbf{X}_k \in \mathbb{R}^{|\mathcal{V}_k| \times d_x}$ denotes the collection of node features. Following the setting in [57], each node $i \in \mathcal{V}_k$ is represented by a feature vector $\mathbf{x}_{k,i} \in \mathbf{X}_k$ derived from its textual description using a pretrained language model encoder (e.g., RoBERTa [34]), where d_x corresponds to the output embedding dimension of the encoder. The learning objectives may vary across clients. For instance, some clients perform node classification, while others focus on link prediction. Each client is equipped with a pretrained large language model (LLM). The LLM is kept frozen during training, while a graph projector is introduced to map graph representations into the LLM input space [57]. The goal is to collaboratively optimize this projector across clients, thereby enhancing the LLM's ability to reason over graph-structured data via federated learning.

Formally, let θ denote the shared graph projector parameters to be collaboratively optimized across clients. The global objective is defined as:

$$\min_{\theta} \mathcal{L}(\theta) = \sum_{k=1}^K w_k \mathbb{E}[\mathcal{L}_k(\theta, \mathcal{G}_k)], \quad (5)$$

where $\mathcal{L}_k$ is the task-specific loss on client k , and $\mathbf{w} = [w_1, \dots, w_K]$ are the aggregation weights across clients.

4.2 Framework Overview

The overall framework of FedLGS is illustrated in Figure 2, showing the interaction between the server and clients. FedLGS comprises two main modules: geometry-aware graph projection (GGP) and shared subspace aggregation (SSA). *First*, in GGP, each client maps node features into a product manifold composed of three complementary Riemannian geometries: hyperbolic ($\mathbb{H}$), spherical ($\mathbb{S}$), and Euclidean ($\mathbb{E}$). The graph structure is adaptively refined by integrating precomputed Ollivier-Ricci curvature $\rho(i, j)$ as a prior, which reweights edges $f(\rho(i, j))$ around target nodes and enables geometry-aware relational encoding. Node representations are then learned within these manifolds, capturing both structural and geometric characteristics. Finally, the node representations are projected into the LLM input token space, and the graph projector is optimized under the client's local task objective. *Second*, in SSA, the server collects the updated graph projector and curvature-based importance vector γ from each client. The projector is decomposed into its manifold components. For each manifold $\mathcal{M}_p$, client updates

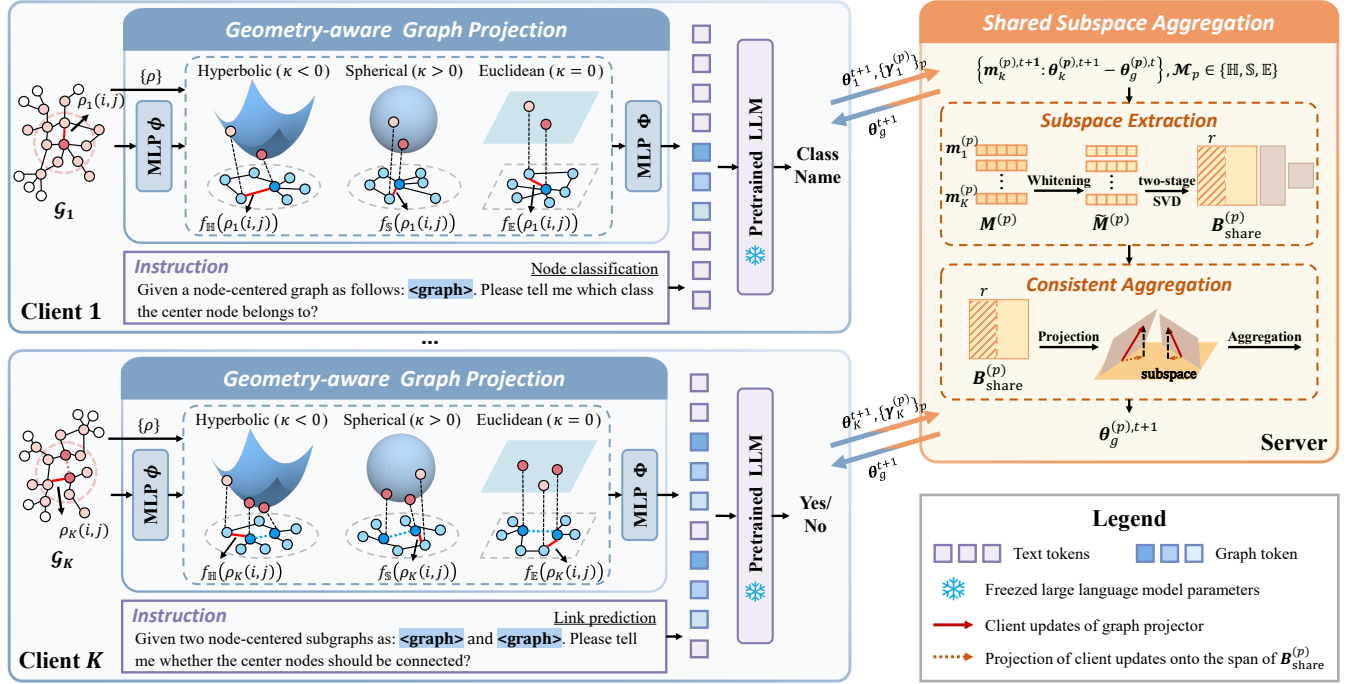


Figure 2: The main framework of the proposed FedLGS.

are first rescaled with γ to ensure consistency across geometries, followed by SVD to extract local subspaces $B_k^{(p)}$. The local subspaces of manifold $\mathcal{M}_p$ are then merged through a second SVD to obtain the shared subspace $B_{\text{share}}^{(p)}$. Global updates are obtained by projecting client updates onto these shared subspaces and averaging the resulting coefficients. The updated projector is then distributed back to clients for the next training round, and this communication process repeats until convergence.

4.3 Geometry-aware Graph Projection

Motivation. Integrating a projector to map graph representations into the LLM input space has shown promising potential. However, existing methods typically rely on a vanilla GNN as the projector [50] or a hand-crafted template [6] to encode structural information. Such approaches struggle to capture the non-Euclidean characteristics of graphs and are further limited in federated settings where labeled data are scarce. To address this, we propose to enhance LLM inputs with geometry-aware graph representations. Specifically, in each client, node representations are mapped into three complementary Riemannian manifolds, which jointly preserve hierarchical, community, and local patterns that are difficult to capture in a single space. Within each manifold, Ollivier–Ricci curvature is incorporated as a prior to guide representation refinement, ensuring that learning is consistent with intrinsic geometric characteristics. The representations are then projected into the LLM input space, offering geometry-aware and semantically enriched tokens.

Product Manifold Projection. To enable geometry-aware graph encoding, we construct a product manifold [14] that integrates complementary geometric components. For each client k , the product

manifold is defined as:

$$\mathbb{P} = \times_{p=1}^{\mathcal{P}} \mathcal{M}_p^{\kappa_p, d_m} = \mathbb{H}^{\kappa_h, d_m} \times \mathbb{S}^{\kappa_s, d_m} \times \mathbb{E}^{\kappa_e=0, d_m}, \quad (6)$$

where $\mathcal{P}$ is the number of manifolds, and κ_p represents the curvature of the p -th manifold. $\kappa_h < 0$ and $\kappa_s > 0$ are learnable curvature parameters (initialized as $\kappa_h = -1$ and $\kappa_s = +1$), while $\kappa_e = 0$ corresponds to the flat Euclidean space. Given the initial node features $X_k \in \mathbb{R}^{|\mathcal{V}_k| \times d_x}$, each manifold component learns its own transformation $\phi_p : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_m}$ to generate a geometry-specific latent representation. The transformed features are then projected onto the product manifold using the exponential map at the origin:

$$Z_k = \left\|_{p=1}^{\mathcal{P}} \exp_0^{\kappa_p} \left(\phi_p(X_k) \right), \quad (7)$$

where $\exp_0^{\kappa_p}(\cdot)$ denotes the exponential map from the tangent space at the origin onto $\mathcal{M}_p$. This formulation ensures that the projected features Z_k inherit the inductive biases of each manifold, allowing the model to capture complementary geometric patterns beyond Euclidean space.

Learning on Manifolds. With the initial embeddings Z_k placed on the product manifold, node representations are refined through a geometry-aware learning process that incorporates edge curvature priors. The key idea is to transform the precomputed Ollivier–Ricci curvature $\rho_k(i, j)$ into geometry-sensitive coefficients that modulate edge importance, allowing each manifold to interpret relational structures in accordance with its intrinsic geometry. For each edge (i, j) on manifold $\mathcal{M}_p$, the curvature-adjusted logit is defined as:

$$s_{ij}^{(p)} = \ln \widehat{\omega}_{ij} + \psi^{(p)}(\rho_k(i, j)), \quad (8)$$

where $\widehat{\omega}_{ij}$ is the base edge weight (for unweighted graphs, $\widehat{\omega}_{ij} = 1/|\mathcal{N}(i)|$). The natural logarithmic transformation converts the base weight into logit space, enhancing numerical stability and

facilitating integration with softmax-based aggregation. The function $\psi^{(p)}(\cdot)$ is a manifold-specific learnable neural network that adaptively maps curvature values into additive corrections, thereby allowing different manifolds to interpret edge curvature in distinct geometric contexts. The normalized propagation weights are obtained by applying a per-destination softmax:

$$f_{M_p}(\rho_k(i, j)) = \frac{\text{Exp}\left(s_{ij}^{(p)}\right)}{\sum_{j' \in N(i)} \text{Exp}\left(s_{ij'}^{(p)}\right)}, \quad (9)$$

which ensures that incoming edges to each node i form a valid probability distribution. Here $\text{Exp}(\cdot)$ denotes the standard real exponential function. Given these geometry- and curvature-aware weights, the update of node i on manifold M_p proceeds in two steps. Let $z_i^{(p)} \in M_p$ denote the representation of node i on manifold M_p . First, neighbor embeddings are aggregated in the tangent space at $z_i^{(p)}$:

$$v_i^{(p)} = \sum_{j \in N(i)} f_{M_p}(\rho_k(i, j)) \log_{z_i^{(p)}}^{\kappa_p} \left(z_j^{(p)} \right), \quad (10)$$

where $\log_z^{\kappa_p}$ denotes the Riemannian logarithmic map at base point z . Then, the aggregated tangent vector is mapped back to the manifold via the exponential map:

$$z_i^{(p)} = \exp_{z_i^{(p)}}^{\kappa_p} \left(v_i^{(p)} \right). \quad (11)$$

By iteratively applying this update (twice as default), each manifold component refines embeddings that account for local connectivity, guided by edge curvature under the intrinsic geometry of the manifold.

Graph-to-Token Projector Optimization. Following iterative refinements on the product manifold, the resulting node representations are prepared for integration with the LLM. For each manifold component, the final embedding $z_i^{(p)}$ is first mapped back to the tangent space at the origin via the logarithmic map, and then transformed through a manifold-specific neural projector to align with the LLM embedding space:

$$h_i^{(p)} = \Phi^{(p)} \left(\log_0^{\kappa_p} \left(z_i^{(p)} \right) \right), \quad (12)$$

where $\Phi^{(p)}(\cdot)$ denotes a learnable mapping tailored to each manifold. The resulting $h_i^{(p)} \in \mathbb{R}^{d_{\text{LLM}}}$ shares the same dimensionality as the LLM's word embeddings, allowing direct insertion into the model's token sequence while preserving geometry-aware relational structure. Instead of collapsing all components into one vector, the projected tokens are kept distinct, thereby preserving complementary inductive biases from hyperbolic, spherical, and Euclidean geometries. These tokens replace the placeholder <graph> in the prompt, with detailed prompt designs provided in Appendix A.2. The training objective aligns the LLM's predictions with graph-based supervision signals. For node classification tasks, the target corresponds to the categorical name of each node, whereas for link prediction, binary labels (yes/no) indicate the existence of an edge. Given LLM output logits $\mathbf{O} \in \mathbb{R}^{L \times |\mathcal{T}|}$ over the vocabulary $\mathcal{T}$ and the target sequence $Y = [y_1, \dots, y_L]$ of length L , the training loss is defined as the standard cross-entropy:

$$\mathcal{L}_{\text{task}} = -\frac{1}{L} \sum_{l=1}^L \ln \frac{\text{Exp}(o_{l, y_l})}{\sum_{t \in \mathcal{T}} \text{Exp}(o_{l, t})}. \quad (13)$$

Through this process, the geometry-aware projector is optimized to provide the LLM with semantically enriched tokens, enabling the LLM to interpret graphs more effectively.

4.4 Shared Subspace Aggregation

Motivation. Many existing FGL methods are primarily designed for node classification [18, 25] and thus struggle to generalize when clients engage in heterogeneous tasks. Some alternative approaches aggregate clients based on pairwise similarity [2, 27], but these strategies often fail to account for geometric inductive biases, since different clients may emphasize distinct semantic aspects of the data. To address this, we propose to extract geometry-consistent update directions shared across clients. Specifically, projector updates are first whitened with curvature-weighted scaling to balance dimension contributions under the intrinsic geometry. The normalized updates are decomposed by a two-stage SVD, where client-specific subspaces are first extracted and then combined into a global shared subspace. Finally, aggregation is performed within this shared subspace, ensuring that the global projector generalizes effectively across heterogeneous client tasks.

Curvature-weighted Whitening. After communication round t , each client k uploads its updated graph projector parameters θ_k^{t+1} , along with a curvature-based importance vector $\gamma_k^{(p)}$ for each manifold $M_p \in \{\mathbb{H}, \mathbb{S}, \mathbb{E}\}$. This vector captures how edge curvatures influence projector parameters during local training, and is computed as:

$$\gamma_{k,d}^{(p)} = \frac{\sum_{(i,j) \in \mathcal{E}_k} |\rho_k(i, j)| \left\| \frac{\partial}{\partial \theta_{k,d}^{(p)}} \frac{1}{2} \|h_i^{(p)} - h_j^{(p)}\|_2^2 \right\|_2}{\sum_{(i,j) \in \mathcal{E}_k} \left\| \frac{\partial}{\partial \theta_{k,d}^{(p)}} \frac{1}{2} \|h_i^{(p)} - h_j^{(p)}\|_2^2 \right\|_2 + \varepsilon_1}, \quad (14)$$

where $d = 1, \dots, D_p$ and D_p denotes the total number of projector parameters associated with manifold M_p . ε_1 is a small positive constant for numerical stability. This formulation measures how sensitively each parameter dimension responds to curvature-weighted variations in the pairwise node representations, thereby quantifying the geometric influence of local training dynamics. For each manifold, the client update vectors are defined as:

$$\mathbf{m}_k^{(p), t+1} = \text{vec} \left(\theta_k^{(p), t+1} - \theta_g^{(p), t} \right) \in \mathbb{R}^{D_p}. \quad (15)$$

For simplicity of presentation, we omit the round index $t + 1$ in the following derivations and denote the update vector simply as $\mathbf{m}_k^{(p)}$. Stacking all client updates gives the update matrix $\mathbf{M}^{(p)} = [\mathbf{m}_1^{(p)}, \dots, \mathbf{m}_K^{(p)}] \in \mathbb{R}^{D_p \times K}$. Directly decomposing $\mathbf{M}^{(p)}$ is suboptimal, because parameter dimensions exhibit uneven variances, and geometry-relevant directions may be distorted and dominated by high-variance components. To correct for these imbalances, we introduce a geometry-aware whitening. Specifically, each parameter dimension is normalized by its empirical variance across clients and further weighted by curvature-derived importance. The diagonal variance matrix $\text{Var}^{(p)} \in \mathbb{R}^{D_p \times D_p}$ is:

$$\text{Var}^{(p)} = \text{diag} \left(\frac{1}{K} \mathbf{M}^{(p)} \mathbf{M}^{(p)\top} \right). \quad (16)$$

Using $\text{Var}^{(p)}$ and the curvature-based importance vector $\gamma^{(p)}$ by averaging the $\gamma_k^{(p)}$ over clients, we define the geometry-aware scaling matrix as:

$$G^{(p)} = \text{diag} \left(\frac{\gamma^{(p)}}{\text{Var}^{(p)} + \varepsilon_2} \right), \quad (17)$$

with $\varepsilon_2 > 0$ for stability. This construction downscales directions of high empirical variance while amplifying dimensions associated with strong curvature signals, thereby aligning parameter updates with geometry-consistent importance. To further balance contributions across clients, we reweight columns by:

$$\alpha_k \propto \left\| \left(G^{(p)} \right)^{1/2} \mathbf{m}_k^{(p)} \right\|_2^{-2}, \quad \sum_{k=1}^K \alpha_k = K, \quad (18)$$

ensuring that disproportionately large updates do not dominate. The whitened update matrix is then defined as:

$$\tilde{\mathbf{M}}^{(p)} = \left(G^{(p)} \right)^{1/2} \mathbf{M}^{(p)} \text{diag}(\sqrt{\alpha}), \quad (19)$$

from which geometry-consistent directions can be extracted. More details are provided in Appendix G.

Shared Subspace Extraction. The server then performs a two-stage singular value decomposition (SVD) [54]. For client k , the whitened update vector is $\tilde{\mathbf{m}}_k^{(p)} = \tilde{\mathbf{M}}_{[:,k]}^{(p)} \in \mathbb{R}^{D_p}$. The set of parameter names on manifold $\mathcal{M}_p$ is written as $\mathcal{S}_p$. For each $\ell \in \mathcal{S}_p$, the associated index set is $\mathcal{I}_\ell \subset \{1, \dots, D_p\}$ with size $D_{p,\ell} = |\mathcal{I}_\ell|$. The slice of $\tilde{\mathbf{m}}_k^{(p)}$ indexed by $\mathcal{I}_\ell$ is $\tilde{\mathbf{m}}_{k,\ell}^{(p)} = \tilde{\mathbf{M}}_{[\mathcal{I}_\ell,k]}^{(p)} \in \mathbb{R}^{D_{p,\ell}}$, which is reshaped to its native matrix form $\Delta \tilde{\mathbf{W}}_{k,\ell}^{(p)} = \text{reshape}(\tilde{\mathbf{m}}_{k,\ell}^{(p)}) \in \mathbb{R}^{d_\ell^{\text{out}} \times d_\ell^{\text{in}}}$, so that $D_{p,\ell} = d_\ell^{\text{out}} d_\ell^{\text{in}}$. For each ℓ , perform SVD on the client-specific matrix:

$$\Delta \tilde{\mathbf{W}}_{k,\ell}^{(p)} = \mathbf{U}_{k,\ell}^{(p)} \Sigma_{k,\ell}^{(p)} (\mathbf{V}_{k,\ell}^{(p)})^\top, \quad (20)$$

where $\mathbf{U}_{k,\ell}^{(p)} \in \mathbb{R}^{d_\ell^{\text{out}} \times d_\ell^{\text{out}}}$ and $\mathbf{V}_{k,\ell}^{(p)} \in \mathbb{R}^{d_\ell^{\text{in}} \times d_\ell^{\text{in}}}$. A hyperparameter $r \in (0, 1]$, referred to as the preserved rank ratio, specifies the proportion of singular vectors to retain. The retained rank is $r_\ell = \lfloor r \cdot d_\ell^{\text{out}} \rfloor$, and the first r_ℓ columns of $\mathbf{U}_{k,\ell}^{(p)}$ are taken as the local bases:

$$\mathbf{B}_{k,\ell}^{(p)} = \mathbf{U}_{k,\ell}^{(p)}[:, 1:r_\ell] \in \mathbb{R}^{d_\ell^{\text{out}} \times r_\ell}, \quad (21)$$

which captures the r_ℓ dominant row-space directions of client k for parameter ℓ on manifold $\mathcal{M}_p$ in the whitened space. Since these local bases may overlap but are not fully aligned across clients, a second stage is required to extract a shared space. The server concatenates all local bases to get a new matrix $\mathbf{H}_\ell^{(p)} = [\mathbf{B}_1^{(p)}, \dots, \mathbf{B}_K^{(p)}] \in \mathbb{R}^{d_\ell^{\text{out}} \times (Kr_\ell)}$ and performs another SVD:

$$\mathbf{H}_\ell^{(p)} = \mathbf{U}_{\text{share},\ell}^{(p)} \Sigma_{\text{share},\ell}^{(p)} (\mathbf{V}_{\text{share},\ell}^{(p)})^\top. \quad (22)$$

The shared basis for parameter ℓ is defined by the first r_ℓ columns of $\mathbf{U}_{\text{share},\ell}^{(p)}$:

$$\mathbf{B}_{\text{share},\ell}^{(p)} = \mathbf{U}_{\text{share},\ell}^{(p)}[:, 1:r_\ell] \in \mathbb{R}^{d_\ell^{\text{out}} \times r_\ell}. \quad (23)$$

The span of $\mathbf{B}_{\text{share},\ell}^{(p)}$ defines a subspace that captures geometry-consistent directions across heterogeneous clients. This two-stage process is applied separately for each parameter name ℓ , i.e., we do not mix different tensors when extracting subspaces.

Consistent Aggregation. Once the shared bases $\{\mathbf{B}_{\text{share},\ell}^{(p)}\}_{\ell \in \mathcal{S}_p}$ are extracted for manifold $\mathcal{M}_p$, the server aggregates updates within

these consistent subspaces rather than in the full parameter space. Because $\mathbf{B}_{\text{share},\ell}^{(p)}$ are orthonormal in the whitened metric, the block-diagonal projector in the vectorized space is constructed as:

$$\mathbf{P}_{\text{blk}}^{(p)} = \text{blkdiag} \left(\mathbf{P}_\ell^{(p)} \right)_{\ell \in \mathcal{S}_p} \in \mathbb{R}^{D_p \times D_p}, \quad (24)$$

where each block projects tensor ℓ onto the consistent row subspace:

$$\mathbf{P}_\ell^{(p)} = \mathbf{I}_{d_\ell^{\text{in}}} \otimes \left(\mathbf{B}_{\text{share},\ell}^{(p)} \mathbf{B}_{\text{share},\ell}^{(p)\top} \right). \quad (25)$$

Here $\otimes$ denotes the Kronecker product, so that the row-space projector is replicated across input channels in the vectorized space. However, when aggregating, updates cannot be projected directly with $\mathbf{P}_{\text{blk}}^{(p)}$, because this operator is only valid in the whitened space where parameter scales have been normalized. If it is applied in the original parameter space, directions with large variance would reappear and distort the consistency. To avoid this issue, the G -consistent projection first maps the update into the whitened space, applies the projector, and then transforms the result back:

$$\text{Proj}_{\text{share}}^{(p)}(\mathbf{m}_k^{(p)}) = \left(G^{(p)} \right)^{-1/2} \mathbf{P}_{\text{blk}}^{(p)} \left(G^{(p)} \right)^{1/2} \mathbf{m}_k^{(p)}. \quad (26)$$

This construction ensures that aggregation respects the geometry-corrected scales and retains only the consistent directions that are stable across clients. The server aggregates the projected updates by a weighted average:

$$\bar{\mathbf{m}}^{(p)} = \sum_{k=1}^K w_k \text{Proj}_{\text{share}}^{(p)}(\mathbf{m}_k^{(p)}), \quad (27)$$

where $w_k = \frac{|\mathcal{V}_k|}{\sum_{j=1}^K |\mathcal{V}_j|}$ is the local sample weight of client k . Finally, the global model for manifold $\mathcal{M}_p$ is updated as:

$$\theta_g^{(p),t+1} = \theta_g^{(p),t} + \bar{\mathbf{m}}^{(p)}. \quad (28)$$

By construction, only directions consistently present across clients within the geometry-aware shared subspaces are retained, preventing high-variance spurious dimensions from dominating. The procedure is applied separately to the hyperbolic, spherical, and Euclidean components, and the resulting updates are combined into the new global projector θ_g^{t+1} . The overall algorithm is provided in Appendix H.

5 Experiments

5.1 Experimental Setups

Datasets. We conduct experiments on six widely used graph learning datasets, Pubmed [59], arXiv [17], WikiCS [38], Reddit [19], Photo [40], and Computer [40]. To comprehensively study the federated graph learning scenario, we partition each dataset into three different scales, i.e., 6, 10, and 20 clients, using METIS partitioning following [2]. The clients are evenly divided between two fundamental graph learning tasks. Half of the clients are assigned node classification tasks, and the remaining half perform link prediction tasks. This setup reflects realistic federated environments where different clients may have diverse learning objectives. To simulate the challenging low-supervision scenario common in federated settings, we adopt an extremely sparse labeling strategy. For each client, only 1% of the available data is used for training, while 10% is allocated for validation and another 10% for testing. This limited supervision setting effectively evaluates the ability to learn from scarce labeled data while maintaining robust performance.

Table 1: Comparison of federated graph learning performance.

Datasets		Pubmed						arXiv					
Methods \ # Clients		6		10		20		6		10		20	
		Acc	Macro-F1	ACC	Macro-F1	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1
FedAvg		67.73±0.33	57.23±0.60	67.46±0.34	58.73±0.89	68.08±0.78	54.82±0.64	61.16±0.41	47.28±1.40	61.98±0.28	46.68±0.33	60.35±0.06	42.83±0.93
FedPUB		66.40±0.26	50.75±0.78	68.47±0.35	50.17±0.64	66.59±0.14	46.92±1.32	66.69±0.10	52.10±0.38	66.28±0.09	51.41±0.04	64.63±0.13	50.39±0.19
Fed-RiemannGFM		71.34±0.81	58.58±1.13	70.36±0.53	55.68±1.02	68.25±0.32	48.91±0.53	54.78±0.18	38.12±0.27	54.31±0.43	38.68±1.02	52.86±0.42	37.00±0.04
Fed-ENGINE		67.19±0.58	59.23±0.89	68.81±0.65	57.29±0.58	67.06±1.12	53.06±1.03	65.23±0.62	51.90±1.58	64.42±0.12	48.75±1.17	61.12±1.23	43.44±1.74
Fed-GraphGPT		63.46±1.25	42.79±0.43	63.63±0.35	42.91±1.52	62.63±0.58	39.26±1.38	62.06±1.81	39.70±0.77	61.67±1.98	38.59±1.04	61.23±0.94	38.39±0.38
Fed-LLaGA		75.36±1.05	66.96±1.64	76.81±3.03	67.98±2.19	76.16±3.24	63.70±3.48	69.23±0.34	54.72±2.19	68.04±0.78	54.44±0.25	68.65±0.69	54.65±0.70
FedLGS		79.86±0.31	72.19±0.32	78.93±0.94	71.93±0.75	78.67±0.55	67.17±0.97	73.08±0.43	62.08±0.90	72.98±0.53	61.68±0.86	72.05±0.35	57.64±0.43
Datasets		Reddit						WikiCS					
Methods \ # Clients		6		10		20		6		10		20	
		Acc	Macro-F1	ACC	Macro-F1	Acc	Macro-F1	Acc	Macro-F1	ACC	Macro-F1	Acc	Macro-F1
FedAvg		54.34±0.69	46.82±1.05	56.33±0.60	50.11±3.91	55.26±0.08	49.93±0.16	72.38±0.39	47.73±0.53	66.29±0.29	45.45±0.66	68.33±0.07	46.51±0.11
FedPUB		60.46±0.41	57.98±0.63	61.56±1.08	58.72±0.87	59.29±0.10	54.91±0.24	74.65±0.29	48.92±0.20	68.93±0.21	46.49±0.19	68.69±0.53	44.13±0.57
Fed-RiemannGFM		57.90±0.29	54.30±1.07	54.93±0.33	49.14±0.44	54.70±0.95	48.43±4.15	69.23±0.10	41.70±0.76	67.41±0.93	41.26±1.48	59.57±1.09	39.59±0.85
Fed-ENGINE		58.54±1.74	56.15±2.02	58.59±0.65	56.60±0.38	57.63±0.31	52.78±2.04	63.81±1.12	38.53±0.83	61.83±0.38	36.18±0.19	52.61±0.75	33.32±2.35
Fed-GraphGPT		56.62±1.48	37.92±1.02	56.01±1.24	37.64±0.91	55.21±0.62	37.03±0.03	69.30±0.37	36.64±1.93	63.11±0.14	31.60±2.75	61.28±0.77	34.97±0.13
Fed-LLaGA		57.74±3.22	53.10±4.60	56.49±0.76	52.99±2.93	54.80±2.11	52.03±3.30	75.36±2.01	53.73±3.96	74.88±1.87	50.22±2.43	71.33±2.65	48.41±1.85
FedLGS		64.51±1.21	59.01±1.44	64.39±1.47	58.99±1.61	62.86±1.43	56.32±1.78	81.28±1.22	58.58±1.34	78.21±1.78	55.00±1.04	74.88±0.80	54.80±1.28
Datasets		Photo						Computer					
Methods \ # Clients		6		10		20		6		10		20	
		Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1
FedAvg		67.55±0.21	44.33±0.84	69.83±1.21	53.01±0.06	66.40±0.19	42.94±2.21	69.73±0.18	47.65±0.67	68.96±0.38	49.11±0.90	66.66±0.26	45.30±0.57
FedPUB		69.77±0.03	44.35±0.37	69.90±0.15	43.29±0.11	70.46±0.03	47.43±0.20	72.36±0.38	50.17±0.44	71.75±0.14	50.96±0.24	69.47±0.18	46.27±0.79
Fed-RiemannGFM		69.68±0.31	46.24±1.02	66.90±1.09	43.13±1.06	66.13±0.33	40.86±0.28	72.82±0.21	50.66±0.35	69.02±0.50	47.41±0.67	67.39±0.22	46.47±0.15
Fed-ENGINE		64.24±0.29	40.83±0.51	63.66±0.46	38.41±0.55	61.63±0.65	35.46±1.26	65.15±0.67	40.70±1.60	63.67±0.63	39.34±1.37	58.51±0.47	36.57±1.82
Fed-GraphGPT		69.62±0.18	38.67±0.19	69.26±1.21	37.56±1.53	71.24±0.36	37.72±1.04	76.09±0.49	44.00±0.73	73.07±0.61	40.81±0.49	72.61±0.17	44.89±1.19
Fed-LLaGA		70.47±0.50	49.82±2.87	72.37±2.66	52.41±2.14	75.47±2.09	52.80±2.31	77.07±0.15	55.98±0.91	76.25±0.37	56.32±0.46	74.97±0.45	56.06±0.71
FedLGS		78.42±0.88	56.82±1.25	80.07±1.13	59.35±1.32	79.77±1.04	56.08±1.08	81.24±0.67	61.48±0.89	80.61±0.33	60.34±0.76	79.55±0.76	60.54±0.89

Baselines. To evaluate the effectiveness of FedLGS and demonstrate its advantages, we compare it with the following representative baselines. We first consider two conventional federated learning frameworks: (1) FedAvg [37], the standard federated averaging method applied to GNN training; and (2) FedPUB [2], a personalized subgraph federated learning method with similarity-based weighted aggregation. We further extend recent graph foundation models to the federated setting: (3) Fed-RiemannGFM [46], which adopts Riemannian geometry to learn node representations; (4) Fed-ENGINE [67], which employs an LLM as an enhancer for GNN via lightweight structural adapters; and (5) Fed-GraphGPT [50] and (6) Fed-LLaGA [6], both using LLMs as predictors with a learnable graph-to-token projector collaboratively trained across clients for structural and linguistic alignment.

Implementation Details. We employ Ministral-8B-Instruct-2410 [39] as the base large language model for all compared methods. Additional experiments with different large language models are provided in Appendix D. All models are optimized using the AdamW optimizer, with the learning rate tuned within the range of 1×10^{-2} to 1×10^{-5} . Each client performs one local epoch per communication round, and the total number of rounds is set to 100. The batch size is fixed at 8, manifold component dimensionality is 512, and the preserved rank ratio defaults to 0.3. Model performance is evaluated using both accuracy and Macro-F1, and the reported test results correspond to the checkpoint achieving the best validation performance.

5.2 Performance Evaluation

Main Results. Table 1 shows the average performance results and standard deviation obtained with different seeds. We have the following observations: (1) Conventional federated learning and personalized subgraph-based methods struggle to handle heterogeneous tasks across clients. An exception is observed for FedPUB on the Reddit dataset, possibly due to its binary classification setup being closer in structure to link prediction tasks, resulting in more stable similarity-based aggregation. (2) Using Riemannian manifolds to extract node features achieves better results than the vanilla GNN. RiemannGFM demonstrates strong performance on most datasets, showing the benefit of geometry-aware representation learning. However, its performance on arXiv and WikiCS is relatively weaker, likely because the model’s representation extraction partly relies on pretraining, and these two datasets differ substantially from the pretraining sources. This observation further indicates that conventional GNNs still suffer from limited generalization ability when facing diverse graph distributions. (3) Leveraging LLMs as enhancers does not consistently improve performance. For instance, Fed-ENGINE underperforms in most cases, highlighting the limited adaptability of such approaches to heterogeneous tasks. (4) Employing LLMs as predictors shows promising results. Fed-LLaGA serves as the runner-up across most datasets, benefiting from its unified graph-to-token alignment strategy. In contrast, Fed-GraphGPT is more sensitive to task heterogeneity due to its multi-stage training pipeline, where the intermediate representation mismatch between

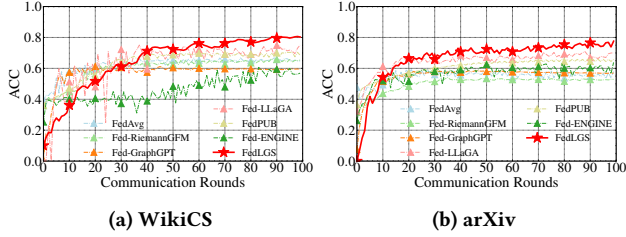


Figure 3: Comparison of convergence.

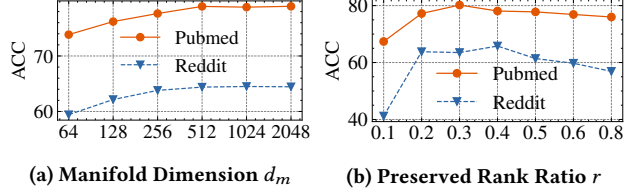


Figure 4: Effect of hyperparameters.

graph encoders and language models may amplify local task discrepancies. (5) FedLGS consistently outperforms all other methods across various scenarios. Moreover, its performance variance remains well controlled, demonstrating the stability and effectiveness.

Convergence Comparison. Figure 3 compares the convergence behavior of FedLGS with baseline methods. The figure illustrates the validation accuracy on the WikiCS and arXiv datasets, showing how each method evolves throughout training. Most baselines exhibit noticeable fluctuations in their convergence curves, highlighting the inherent instability of federated graph learning under task heterogeneity. Fed-ENGINE and Fed-LLaGA exhibit even greater variability, indicating that directly integrating LLMs can intensify inconsistencies across clients. In contrast, FedLGS initially converges slightly slower than other baselines, as it requires additional adaptation to refine manifold curvature and stabilize geometry-aware node representations. As training proceeds, it consistently outperforms all baselines. These improvements are attributed to the geometry-aware, semantically enriched LLM inputs and the shared subspace aggregation strategy, which together promote stable and superior performance across heterogeneous tasks.

5.3 In-depth Analysis

Ablation Study. To evaluate the effectiveness of each module in FedLGS, we conduct an ablation study using four datasets, Pubmed, WikiCS, Reddit, and Photo, as shown in Table 2. Two variants are compared against Fed-LLaGA, which directly applies LLMs without geometry-aware or subspace-based designs. The first variant FedLGS-w/o-GGP excludes the geometry-aware graph projection module, replacing the product manifold encoder with a Euclidean GNN projector. This variant verifies whether incorporating geometric representations enhances LLM interpretation and downstream performance. The second version FedLGS-w/o-SSA removes the shared subspace projection module, which aggregates the parameters of the projector using the conventional federated averaging aggregation process. This variant helps evaluate the importance of extracting geometry-consistent update directions across heterogeneous clients. The results in Table 2 clearly demonstrate the

Table 2: Ablation study.

Datasets	Pubmed		WikiCS	
Methods	ACC	Macro-F1	ACC	Macro-F1
Fed-LLaGA	76.81 $\pm$ 3.03	67.98 $\pm$ 2.19	74.88 $\pm$ 1.87	50.22 $\pm$ 2.43
FedLGS -w/o-GGP	77.25 $\pm$ 1.12	69.34 $\pm$ 1.08	76.42 $\pm$ 1.35	52.67 $\pm$ 1.26
FedLGS w/o-SSA	76.89 $\pm$ 1.24	68.75 $\pm$ 1.17	75.83 $\pm$ 1.42	51.92 $\pm$ 1.31
FedLGS	78.93$\pm$0.94	71.93$\pm$0.75	78.21$\pm$1.78	55.00$\pm$1.04

Datasets	Reddit		Photo	
Methods	ACC	Macro-F1	ACC	Macro-F1
Fed-LLaGA	56.49 $\pm$ 0.76	52.99 $\pm$ 2.93	72.37 $\pm$ 2.66	52.41 $\pm$ 2.14
FedLGS -w/o-GGP	60.35 $\pm$ 0.89	57.28 $\pm$ 1.42	77.56 $\pm$ 1.28	56.89 $\pm$ 1.21
FedLGS w/o-SSA	59.74 $\pm$ 0.94	56.63 $\pm$ 1.48	76.84 $\pm$ 1.35	55.72 $\pm$ 1.27
FedLGS	64.39$\pm$1.47	58.99$\pm$1.61	80.07$\pm$1.13	59.35$\pm$1.32

effectiveness of both modules, validating their individual and combined contributions to the overall performance. Notably, FedLGS-w/o-SSA exhibits a more significant performance drop, highlighting the critical role of the shared subspace aggregation in stabilizing cross-client updates and preserving geometry-consistent directions under graph task heterogeneity.

Hyperparameter Sensitivity. To investigate the sensitivity of hyperparameters, we conduct experiments on Pubmed and Reddit. We tune the per manifold dimension $d_m = \{64, 128, 256, 512, 1024, 2048\}$, and preserved rank ratio $r = \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8\}$. As shown in Figure 4, the performance generally improves as the manifold dimension d_m increases, since higher dimensions enable richer geometry and more expressive representations. However, the improvement becomes marginal beyond $d_m = 512$, indicating that overly high dimensions may introduce redundancy without further benefiting the embedding quality. In contrast, the preserved rank ratio r exhibits a bell-shaped trend. Both too small and too large values of r degrade accuracy, as excessive compression removes essential geometry-aware components while overly relaxed preservation introduces noise. Empirically, r values between 0.3 and 0.5 achieve the best balance between stability and performance.

6 Conclusion

In this work, we propose FedLGS, a novel LLM-enhanced federated graph learning framework designed to handle heterogeneous client tasks. FedLGS contains two modules, i.e., geometry-aware graph projection (GGP) and shared subspace aggregation (SSA). GGP embeds local graph representations into complementary Riemannian manifolds guided by Ollivier–Ricci curvature, enabling geometry-aware and semantically enriched inputs that enhance LLM interpretation with limited supervision. Meanwhile, SSA applies curvature-weighted whitening and subspace-based aggregation to align heterogeneous client updates within geometry-consistent directions, thus achieving robust and generalizable global optimization. Extensive experiments across six benchmarks demonstrate that FedLGS consistently improves FGL performance under task heterogeneity.

Acknowledgments

This work was supported by the National Key R&D Program of China (2022YFB4501500, 2022YFB4501504), and the National Natural Science Foundation of China (62522217).

References

- [1] Gregor Bachmann, Gary Bécigneul, and Octavian Ganea. 2020. Constant curvature graph convolutional networks. In *International conference on machine learning*. PMLR, 486–496.
- [2] Jinheon Baek, Wonyong Jeong, Jiongdao Jin, Jaehong Yoon, and Sung Ju Hwang. 2023. Personalized subgraph federated learning. In *International Conference on Machine Learning*. PMLR, 1396–1415.
- [3] Mikhail Belkin and Partha Niyogi. 2003. Laplacian eigenmaps for dimensionality reduction and data representation. *Neural computation* 15, 6 (2003), 1373–1396.
- [4] Chaochao Chen, Xiaohua Feng, Yuyuan Li, Lingjuan Lyu, Jun Zhou, Xiaolin Zheng, and Jianwei Yin. 2024. Integration of large language models and federated learning. *Patterns* 5, 12 (2024).
- [5] Junze Chen, Cheng Yang, Shujie Li, Zhiqiang Zhang, Yawen Li, Junping Du, and Chuan Shi. 2025. GraphLAMA: Enabling Efficient Adaptation of Graph Language Models with Limited Annotations. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*. 191–202.
- [6] Runjin Chen, Tong Zhao, Ajay Kumar Jaiswal, Neil Shah, and Zhangyang Wang. 2024. LLaGA: Large Language and Graph Assistant. In *International Conference on Machine Learning*. PMLR, 7809–7823.
- [7] Zhikai Chen, Haitao Mao, Hang Li, Wei Jin, Hongzhi Wen, Xiaochi Wei, Shuaiqiang Wang, Dawei Yin, Wenqi Fan, Hui Liu, et al. 2024. Exploring the potential of large language models (llms) in learning on graphs. *ACM SIGKDD Explorations Newsletter* 25, 2 (2024), 42–61.
- [8] Wenqi Fan, Yao Ma, Qing Li, Yuan He, Eric Zhao, Jiliang Tang, and Dawei Yin. 2019. Graph neural networks for social recommendation. In *The world wide web conference*. 417–426.
- [9] Yi Fang, Dongzhe Fan, Daochen Zha, and Qiaoyu Tan. 2024. Gaugllm: Improving graph contrastive learning for text-attributed graphs with large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 747–758.
- [10] Xingbo Fu, Zihan Chen, Yinhan He, Song Wang, Binchi Zhang, Chen Chen, and Jundong Li. 2025. Virtual Nodes Can Help: Tackling Distribution Shifts in Federated Graph Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 16657–16665.
- [11] Xingbo Fu, Zihan Chen, Binchi Zhang, Chen Chen, and Jundong Li. 2024. Federated graph learning with structure proxy alignment. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 827–838.
- [12] Xingbo Fu, Binchi Zhang, Yushun Dong, Chen Chen, and Jundong Li. 2022. Federated graph machine learning: A survey of concepts, techniques, and applications. *ACM SIGKDD Explorations Newsletter* 24, 2 (2022), 32–47.
- [13] Hang Gao, Huang Wenxuan, Fengge Wu, Zhao Junsuo, Changwen Zheng, and Huaping Liu. 2025. LLM Enhancers for GNNs: An Analysis from the Perspective of Causal Mechanism Identification. In *Forty-second International Conference on Machine Learning*. <https://openreview.net/forum?id=UWTz4ai3FZ>
- [14] Albert Gu, Frederic Sala, Beliz Gunel, and Christopher Ré. 2019. Learning Mixed-Curvature Representations in Product Spaces. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=HJxeWnCcF7>
- [15] Xiaoxin He, Xavier Bresson, Thomas Laurent, Adam Perold, Yann LeCun, and Bryan Hooi. 2024. Harnessing Explanations: LLM-to-LM Interpreter for Enhanced Text-Attributed Graph Representation Learning. In *The Twelfth International Conference on Learning Representations*. <https://openreview.net/forum?id=RXFVcynVe1>
- [16] Yufei He, Yuan Sui, Xiaoxin He, and Bryan Hooi. 2025. Unigraph: Learning a unified cross-domain foundation model for text-attributed graphs. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*. 448–459.
- [17] Weihua Hu, Matthias Fey, Marinka Zitnik, Yuxiao Dong, Hongyu Ren, Bowen Liu, Michele Catasta, and Jure Leskovec. 2020. Open graph benchmark: Datasets for machine learning on graphs. *Advances in neural information processing systems* 33 (2020), 22118–22133.
- [18] Wenke Huang, Guancheng Wan, Mang Ye, and Bo Du. 2023. Federated Graph Semantic and Structural Learning. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23*, Edith Elkind (Ed.). International Joint Conferences on Artificial Intelligence Organization, 3830–3838. doi:10.24963/ijcai.2023/426 Main Track.
- [19] Xuanwen Huang, Kaiqiao Han, Yang Yang, Dezheng Bao, Quanjin Tao, Ziwei Chai, and Qi Zhu. 2024. Can gnn be good adapter for llms?. In *Proceedings of the ACM Web Conference 2024*. 893–904.
- [20] Bowen Jin, Gang Liu, Chi Han, Meng Jiang, Heng Ji, and Jiawei Han. 2024. Large language models on graphs: A comprehensive survey. *IEEE Transactions on Knowledge and Data Engineering* (2024).
- [21] Jürgen Jost and Shiping Liu. 2014. Ollivier’s Ricci curvature, local clustering and curvature-dimension inequalities on graphs. *Discrete & Computational Geometry* 51, 2 (2014), 300–322.
- [22] Minku Kang and Hognu Park. 2025. Curriculum guided personalized subgraph federated learning. *arXiv preprint arXiv:2509.00402* (2025).
- [23] Zixuan Ke, Yifei Ming, and Shafiq Joty. 2025. Adaptation of Large Language Models. In *Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 5: Tutorial Abstracts)*. 30–37.
- [24] Agnan Kessy, Alex Lewin, and Korbinian Strimmer. 2018. Optimal whitening and decorrelation. *The American Statistician* 72, 4 (2018), 309–314.
- [25] Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, and Chanyoung Park. 2025. Subgraph Federated Learning for Local Generalization. In *The Thirteenth International Conference on Learning Representations*. <https://openreview.net/forum?id=cH65nS5sOz>
- [26] Lecheng Kong, Jiarui Feng, Hao Liu, Chengsong Huang, Jiaxin Huang, Yixin Chen, and Muhan Zhang. 2025. GOFA: A Generative One-For-All Model for Joint Graph Language Modeling. In *The Thirteenth International Conference on Learning Representations*. <https://openreview.net/forum?id=mljblC9hfm>
- [27] Xunkai Li, Zhengyu Wu, Wentao Zhang, Yinlin Zhu, Rong-Hua Li, and Guoren Wang. 2023. FedGTA: Topology-Aware Averaging for Federated Graph Learning. *Proceedings of the VLDB Endowment* 17, 1 (2023), 41–50.
- [28] Xinting Liao, Weiming Liu, Jiaming Qian, Pengyang Zhou, Jiahe Xu, Wenjie Wang, Chaochao Chen, Xiaolin Zheng, and Tat-Seng Chua. 2025. FOCOOp: Enhancing Out-of-Distribution Robustness in Federated Prompt Learning for Vision-Language Models. In *Forty-second International Conference on Machine Learning*. <https://openreview.net/forum?id=XCLZgmb990>
- [29] Xinting Liao, Weiming Liu, Pengyang Zhou, Fengyuan Yu, Jiahe Xu, Jun Wang, Wenjie Wang, Chaochao Chen, and Xiaolin Zheng. 2024. Foogd: Federated collaboration for both out-of-distribution generalization and detection. *Advances in Neural Information Processing Systems Systems* 37 (2024), 132908–132945.
- [30] Tong Lin and Hongbin Zha. 2008. Riemannian manifold learning. *IEEE transactions on pattern analysis and machine intelligence* 30, 5 (2008), 796–809.
- [31] Hao Liu, Jiarui Feng, Lecheng Kong, Ningyue Liang, Dacheng Tao, Yixin Chen, and Muhan Zhang. 2024. One For All: Towards Training One Graph Model For All Classification Tasks. In *The Twelfth International Conference on Learning Representations*. <https://openreview.net/forum?id=4IT2pgc9v6>
- [32] Rui Liu, Pengwei Xing, Zichao Deng, Anran Li, Cuntai Guan, and Han Yu. 2024. Federated graph neural networks: Overview, techniques, and challenges. *IEEE transactions on neural networks and learning systems* (2024).
- [33] Weiming Liu, Chaochao Chen, Xinting Liao, Mengling Hu, Jianwei Yin, Yan-chao Tan, and Longfei Zheng. 2023. Federated Probabilistic Preference Distribution Modelling with Compactness Co-Clustering for Privacy-Preserving Multi-Domain Recommendation. In *IJCAI*. 2206–2214.
- [34] Yinhan Liu, Mylène Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692* (2019).
- [35] Sichun Luo, Yuanzhang Xiao, Xinyi Zhang, Yang Liu, Wenbo Ding, and Linqi Song. 2024. Perfedrec++: Enhancing personalized federated recommendation with self-supervised pre-training. *ACM Transactions on Intelligent Systems and Technology* 15, 5 (2024), 1–24.
- [36] Xingyuan Mao, Yuwen Liu, Lianying Qi, Li Duan, Xiaolong Xu, Xuyun Zhang, Wanchun Dou, Amin Beheshti, and Xiaokang Zhou. 2024. Cluster-driven personalized federated recommendation with interest-aware graph convolution network for multimedia. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 5614–5622.
- [37] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*. PMLR, 1273–1282.
- [38] Péter Mernyei and Cătălina Cangea. 2020. Wiki-cs: A wikipedia-based benchmark for graph neural networks. *arXiv preprint arXiv:2007.02901* (2020).
- [39] Mistral AI. 2024. Un Ministral, des Ministraux. <https://mistral.ai/news/ministraux>
- [40] Jianmo Ni, Jiacheng Li, and Julian McAuley. 2019. Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*. 188–197.
- [41] Yann Ollivier. 2007. Ricci curvature of metric spaces. *Comptes Rendus Mathématique* 345, 11 (2007), 643–646.
- [42] Chenglu Pan, Jiarong Xu, Yue Yu, Ziqi Yang, Qingbiao Wu, Chunping Wang, Lei Chen, and Yang Yang. 2024. Towards fair graph federated learning via incentive mechanisms. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 14499–14507.
- [43] Qiying Pan, Ruofan Wu, Tianyi Zhang, Yifei Zhu, Weiqiang Wang, et al. [n. d.]. FedGKD: Unleashing the Power of Collaboration in Federated Graph Neural Networks. In *NeurIPS 2023 Workshop: New Frontiers in Graph Learning*.
- [44] Benedetto Piccoli and Francesco Rossi. 2016. On properties of the generalized Wasserstein distance. *Archive for Rational Mechanics and Analysis* 222, 3 (2016), 1339–1365.
- [45] Jiajie Su, Chaochao Chen, Weiming Liu, Zibin Lin, Shuheng Shen, Weiqiang Wang, and Xiaolin Zheng. 2024. Revisit targeted model poisoning on federated recommendation: Optimize via multi-objective transport. In *Proceedings of the 47th international acm sigir conference on research and development in information*

- retrieval. 1722–1732.
- [46] Li Sun, Zhenhao Huang, Suyang Zhou, Qiqi Wan, Hao Peng, and Philip Yu. 2025. Riemanngm: Learning a graph foundation model from riemannian geometry. In *Proceedings of the ACM on Web Conference 2025*. 1154–1165.
 - [47] Yue Tan, Yixin Liu, Guodong Long, Jing Jiang, Qinghua Lu, and Chengqi Zhang. 2023. Federated learning on non-iid graphs via structural knowledge sharing. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 37. 9953–9961.
 - [48] Zihan Tan, Guancheng Wan, Wenke Huang, and Mang Ye. 2024. Fedssp: Federated graph learning with spectral knowledge and personalized preference. *Advances in Neural Information Processing Systems* 37 (2024), 34561–34581.
 - [49] Huayi Tang and Yong Liu. 2023. Towards understanding generalization of graph neural networks. In *International Conference on Machine Learning*. PMLR, 33674–33719.
 - [50] Jiabin Tang, Yuhao Yang, Wei Wei, Lei Shi, Lixin Su, Suqi Cheng, Dawei Yin, and Chao Huang. 2024. Graphgpt: Graph instruction tuning for large language models. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 491–500.
 - [51] Shukichi Tanno. 1988. Ricci curvatures of contact Riemannian manifolds. *Tohoku Mathematical Journal, Second Series* 40, 3 (1988), 441–448.
 - [52] Paul Voigt and Axel Von dem Bussche. 2017. The eu general data protection regulation (gdpr). *A Practical Guide, 1st Ed., Cham: Springer International Publishing* 10, 3152676 (2017), 10–5555.
 - [53] Duo Wang, Yuan Zuo, Fengzhi Li, and Junjie Wu. 2024. Lms as zero-shot graph learners: Alignment of gnn representations with llm token embeddings. *Advances in Neural Information Processing Systems* 37 (2024), 5950–5973.
 - [54] Yongxian Wei, Anke Tang, Li Shen, Zixuan Hu, Chun Yuan, and Xiaochun Cao. 2025. Modeling Multi-Task Model Merging as Adaptive Projective Gradient Descent. In *Forty-second International Conference on Machine Learning*. <https://openreview.net/forum?id=EqoKRSR5Pa>
 - [55] Chuhan Wu, Fangzhao Wu, Lingjuan Lyu, Tao Qi, Yongfeng Huang, and Xing Xie. 2022. A federated graph neural network framework for privacy-preserving personalization. *Nature Communications* 13, 1 (2022), 3091.
 - [56] Shiwen Wu, Fei Sun, Wentao Zhang, Xu Xie, and Bin Cui. 2022. Graph neural networks in recommender systems: a survey. *Comput. Surveys* 55, 5 (2022), 1–37.
 - [57] Xixi Wu, Yifei Shen, Fangzhou Ge, Caihua Shan, Yizhu Jiao, Xiangguo Sun, and Hong Cheng. 2025. When Do LLMs Help With Node Classification? A Comprehensive Analysis. In *Forty-second International Conference on Machine Learning*. <https://openreview.net/forum?id=O3WqAhxuc7>
 - [58] Han Xie, Jing Ma, Li Xiong, and Carl Yang. 2021. Federated graph classification over non-iid graphs. *Advances in Neural Information Processing Systems* 34 (2021), 18839–18852.
 - [59] Zhilin Yang, William Cohen, and Ruslan Salakhudinov. 2016. Revisiting semi-supervised learning with graph embeddings. In *International conference on machine learning*. PMLR, 40–48.
 - [60] Ziqi Yang, Zhaopeng Peng, Zihui Wang, Jianzhong Qi, Chaochao Chen, WeiKe Pan, Chenglu Wen, Cheng Wang, and Xiaoliang Fan. 2024. Federated Graph Learning for Cross-Domain Recommendation. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. <https://openreview.net/forum?id=UBpPOqrBKE>
 - [61] Ke Zhang, Carl Yang, Xiaoxiao Li, Lichao Sun, and Siu Ming Yiu. 2021. Sub-graph federated learning with missing neighbor generation. *Advances in Neural Information Processing Systems* 34 (2021), 6671–6682.
 - [62] Xiaolin Zheng, Jiajie Su, Weiming Liu, and Chaochao Chen. 2022. DDGHM: Dual dynamic graph with hybrid metric training for cross-domain sequential recommendation. In *Proceedings of the 30th ACM international conference on multimedia*. 471–481.
 - [63] Pengyang Zhou, Chaochao Chen, Weiming Liu, Xinting Liao, Wenkai Shen, Jiahe Xu, Zhihui Fu, Jun Wang, Wu Wen, and Xiaolin Zheng. 2025. FedGOG: Federated Graph Out-of-Distribution Generalization with Diffusion Data Exploration and Latent Embedding Decorrelation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 22965–22973.
 - [64] Pengyang Zhou, Chaochao Chen, Weiming Liu, Xinting Liao, Fengyuan Yu, Zhihui Fu, Xingyu Lou, Wu Wen, Xiaolin Zheng, and Jun Wang. 2025. FedGF: Enhancing Structural Knowledge via Graph Factorization for Federated Graph Learning. In *Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining*. 448–456.
 - [65] Pengyang Zhou, Chaochao Chen, Weiming Liu, Wenkai Shen, Xinting Liao, Huarong Deng, Zhihui Fu, Jun Wang, Wu Wen, and Xiaolin Zheng. 2025. Joint item embedding dual-view exploration and adaptive local-global fusion for federated recommendation. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 424–434.
 - [66] Yinlin Zhu, Xunkai Li, Zhengyu Wu, Di Wu, Miao Hu, and Rong-Hua Li. 2024. FedTAD: Topology-aware Data-free Knowledge Distillation for Subgraph Federated Learning. In *IJCAI*.
 - [67] Yun Zhu, Yaoke Wang, Haizhou Shi, and Siliang Tang. 2024. Efficient tuning and inference for large language models on textual graphs. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*. 5734–5742.

A Experimental Setups

A.1 Dataset Statistics

We adopt six graph datasets from diverse domains, as summarized in Table 3. Pubmed [59] and arXiv [17] represent academic citation networks, while WikiCS [38] is a web-link network. Reddit [19] belongs to the social domain, and Photo and Computer [40] are from the e-commerce domain. This diversity ensures that our evaluation covers a wide range of structural and semantic characteristics.

Table 3: Dataset statistics.

Dataset	# total nodes	# total edges	# labeled samples per client		
			$K = 6$	$K = 10$	$K = 20$
Pubmed	19,717	88,670	34	20	10
arXiv	169,343	1,166,243	282	169	85
WikiCS	11,701	431,726	20	12	6
Reddit	33,434	302,876	57	34	17
Photo	48,362	873,793	83	50	25
Computer	87,229	1,256,548	149	90	45

A.2 Prompts

The following examples illustrate the prompt templates used for node classification and link prediction, shown here with the Pubmed dataset for clarity. The same design is consistently applied to other datasets following [6, 57].

Node Classification:

Given a node-centered graph: **<graph>**, each node corresponds to a paper about Diabetes. We need to classify the center node into three classes: Experimentally induced diabetes, Type 1 diabetes, or Type 2 diabetes. Please tell me which class the center node belongs to?

Link Prediction:

Given two node-centered subgraphs **<graph>** and **<graph>**, please tell me whether the two center nodes should be connected?

B Task Heterogeneity in FGL

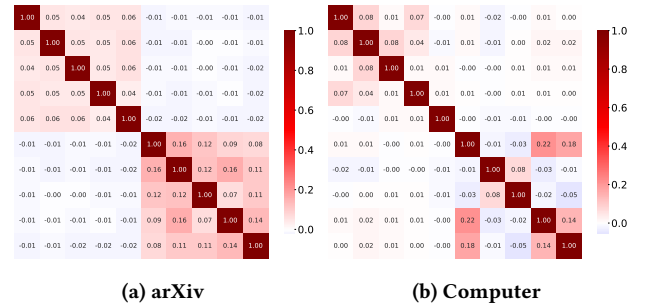


Figure 5: Similarity of client update directions at the 5-th communication round for Fed-LLaGA.

In our setting, clients are assigned different learning tasks. We evaluate 10 clients on the arXiv and Computer datasets, with half

Table 4: Performance under varying heterogeneity.

WikiCS	5:5		7:3		9:1	
Methods	ACC	Macro-F1	ACC	Macro-F1	ACC	Macro-F1
FedPUB	68.93	46.49	73.89	42.09	76.95	38.12
Fed-LLaGA	74.88	50.22	79.08	46.28	72.75	36.73
FedLGS	78.21	55.00	81.46	47.31	81.83	43.13

Photo	5:5		7:3		9:1	
Methods	ACC	Macro-F1	ACC	Macro-F1	ACC	Macro-F1
FedPUB	69.90	43.29	74.93	39.53	77.54	32.30
Fed-LLaGA	72.37	52.41	73.75	44.76	75.00	37.83
FedLGS	80.07	59.35	84.11	56.24	84.94	49.90

Table 5: Integration with various LLMs.

Datasets	Pubmed		WikiCS	
Base Model	ACC	Macro-F1	ACC	Macro-F1
Llama-3.2-3B	77.68±0.89	66.94±0.92	76.92±1.20	53.23±1.15
Qwen3-4B	77.84±1.01	67.57±0.87	73.50±1.35	50.42±1.28
Minstral-8B	78.93±0.94	71.93±0.75	78.21±1.78	55.00±1.04

performing node classification and half link prediction. Heterogeneity is quantified by the cosine similarity between update directions, i.e., the difference between each client model at round $t+1$ and the global model at round t . Figure 5 shows the similarity matrices at the 5th round for Fed-LLaGA. On arXiv, clients on the same task exhibit clear intra-task clustering, while cross-task pairs show negative similarity, indicating strong task-level heterogeneity. In contrast, Computer shows weaker separation and intra-task variability, indicating additional heterogeneity that complicates aggregation.

C Varying Heterogeneity Degrees

To evaluate the degree of heterogeneity, we conduct experiments on WikiCS and Photo with 10 clients as shown in Table 4, where the ratio between node classification and link prediction clients is varied from 5:5 to 7:3 and 9:1. As the proportion of node classification clients increases, the overall accuracy tends to improve, since the global model becomes more aligned with the majority-class distributions emphasized by the node classification task. However, this also amplifies the imbalance problem, as the majority classes benefit from higher prediction confidence while minority classes are increasingly underrepresented, leading to degraded performance on them. Consequently, the Macro-F1, which equally weights all classes, consistently decreases. Despite this trade-off, FedLGS achieves superior results across all settings, and its performance advantage becomes more pronounced as the degree of heterogeneity increases.

D Integration with Various LLMs

We further evaluate FedLGS using multiple LLM backbones to assess its adaptability across models of different sizes and architectures. We compare several representative instruction-tuned LLMs, including Llama-3.2-3B-Instruct, Qwen3-4B-Instruct-2507, and Minstral-8B-Instruct-2410, which cover a wide range of parameter scales from lightweight to medium-sized models. Across all settings, FedLGS consistently achieves strong performance, indicating that the proposed geometry-aware graph projection and shared

subspace aggregation can effectively adapt to diverse linguistic representations and reasoning capacities. Even smaller models, such as Llama-3.2-3B yield competitive results, while the larger Minstral-8B offers additional gains, reflecting a favorable balance between scalability and model capability.

E Riemannian Manifold

We summarize several fundamental operations in the κ -stereographic model [1], which provides a unified framework for hyperbolic, spherical, and Euclidean spaces. In this model, the basic algebraic operation is the Möbius addition. For two points $\mathbf{x}, \mathbf{y} \in \mathcal{M}^{\kappa,d}$:

$$\mathbf{x} \oplus_{\kappa} \mathbf{y} = \frac{(1 - 2\kappa\langle \mathbf{x}, \mathbf{y} \rangle - \kappa\|\mathbf{y}\|^2)\mathbf{x} + (1 + \kappa\|\mathbf{x}\|^2)\mathbf{y}}{1 - 2\kappa\langle \mathbf{x}, \mathbf{y} \rangle + \kappa^2\|\mathbf{x}\|^2\|\mathbf{y}\|^2}, \quad (29)$$

which reduces to the standard Euclidean vector addition $\mathbf{x} + \mathbf{y}$ when $\kappa = 0$. Based on this operation, the geodesic distance between $\mathbf{x}$ and $\mathbf{y}$ can be expressed as:

$$d^{\kappa}(\mathbf{x}, \mathbf{y}) = \frac{2}{\sqrt{|\kappa|}} \tan_{\kappa}^{-1} \left(\sqrt{|\kappa|} \|(-\mathbf{x}) \oplus_{\kappa} \mathbf{y}\|_2 \right), \quad (30)$$

which recovers the Euclidean norm $\|\mathbf{x} - \mathbf{y}\|_2$ when $\kappa = 0$. Furthermore, the Riemannian exponential and logarithmic maps provide the correspondence between the manifold and its tangent space. Given a tangent vector $\mathbf{v} \in \mathcal{T}_{\mathbf{x}}\mathcal{M}^{\kappa,d}$, the exponential map projecting it onto the manifold is:

$$\exp_{\mathbf{x}}^{\kappa}(\mathbf{v}) = \mathbf{x} \oplus_{\kappa} \left(\tan_{\kappa} \left(\frac{\sqrt{|\kappa|} \lambda_{\mathbf{x}}^{\kappa} \|\mathbf{v}\|_2}{2} \right) \frac{\mathbf{v}}{\sqrt{|\kappa|} \|\mathbf{v}\|_2} \right), \quad (31)$$

while the inverse logarithmic map is:

$$\log_{\mathbf{x}}^{\kappa}(\mathbf{y}) = \frac{2}{\sqrt{|\kappa|} \lambda_{\mathbf{x}}^{\kappa}} \tan_{\kappa}^{-1} \left(\sqrt{|\kappa|} \|(-\mathbf{x}) \oplus_{\kappa} \mathbf{y}\|_2 \right) \frac{(-\mathbf{x}) \oplus_{\kappa} \mathbf{y}}{\|(-\mathbf{x}) \oplus_{\kappa} \mathbf{y}\|_2}. \quad (32)$$

Together, these operations establish the foundation for optimization, distance computation, and representation learning on manifolds of constant curvature.

F Ollivier-Ricci Curvature

The exact computation of Ollivier-Ricci Curvature by Eq. (4) requires the computation of the Wasserstein distance, which is expensive. Therefore, we adopt the combinatorial approximations based on node degrees and triangle counts [21]. Let v_i and v_j denote the degrees of nodes i and j , and let $\#(i, j)$ denote the number of triangles containing edge (i, j) . Then the curvature is bounded as:

$$\rho(i, j) \geq - \left(1 - \frac{1}{v_i} - \frac{1}{v_j} - \frac{\#(i, j)}{\min(v_i, v_j)} \right)_+ - \left(1 - \frac{1}{v_i} - \frac{1}{v_j} - \frac{\#(i, j)}{\max(v_i, v_j)} \right)_+ + \frac{\#(i, j)}{\max(v_i, v_j)}, \quad (33)$$

$$\rho(i, j) \leq \frac{\#(i, j)}{\max(v_i, v_j)}. \quad (34)$$

Here, $(s)_+ := \max(s, 0)$ denotes the positive part of a scalar s . An efficient estimate can then be obtained by averaging the two bounds:

$$\rho(i, j) = \frac{1}{2} \left(\rho^{\text{lower}}(i, j) + \rho^{\text{upper}}(i, j) \right). \quad (35)$$

G Curvature-Weighted Whitening

The term whitening [24] refers to a transformation that rescales coordinates so that their effective variance becomes comparable.

In the standard case, given updates $\mathbf{M}$ with empirical covariance $\frac{1}{K}\mathbf{M}\mathbf{M}^\top$, whitening uses a transformation $\mathbf{W}$ such that:

$$\frac{1}{K}(\mathbf{W}\mathbf{M})(\mathbf{W}\mathbf{M})^\top \approx \mathbf{I}, \quad (36)$$

ensuring each dimension contributes equally without being dominated by high-variance directions. However, directly applying isotropic whitening would ignore geometry, since dimensions that are strongly tied to graph curvature carry more semantic relevance for the projector. To address this, we replace the uniform rescaling with a geometry-aware diagonal operator:

$$\mathbf{G}^{(p)} = \text{diag}\left(\frac{\gamma^{(p)}}{\text{Var}^{(p)} + \varepsilon}\right), \quad (37)$$

so that high-variance but less informative dimensions are down-weighted, while geometry-sensitive coordinates (large $\gamma_d^{(p)}$) are amplified. The whitened matrix is then:

$$\tilde{\mathbf{M}}^{(p)} = (\mathbf{G}^{(p)})^{\frac{1}{2}} \mathbf{M}^{(p)} \text{diag}(\sqrt{\alpha}), \quad (38)$$

where client weights α_k are determined to equalize the weighted norm of each column under $\mathbf{G}^{(p)}$:

$$E_k = \|(\mathbf{G}^{(p)})^{\frac{1}{2}} \mathbf{m}_k^{(p)}\|_2^2, \quad \alpha_k \propto \frac{1}{E_k}, \quad \sum_{k=1}^K \alpha_k = K. \quad (39)$$

This ensures that no single client dominates due to scale alone. In this formulation, the covariance in the whitened space becomes:

$$\frac{1}{K} \tilde{\mathbf{M}}^{(p)} \tilde{\mathbf{M}}^{(p)\top} = (\mathbf{G}^{(p)})^{\frac{1}{2}} \left(\frac{1}{K} \mathbf{M}^{(p)} \text{diag}(\alpha) \mathbf{M}^{(p)\top} \right) (\mathbf{G}^{(p)})^{\frac{1}{2}}, \quad (40)$$

whose diagonal entries approximate:

$$\left[\frac{1}{K} \tilde{\mathbf{M}}^{(p)} \tilde{\mathbf{M}}^{(p)\top} \right]_{dd} \approx \gamma_d^{(p)}, \quad (41)$$

i.e., after whitening, the effective variance of each dimension reflects its geometric importance.

We also illustrate the effectiveness from a spectral perspective. The left singular vectors of the whitened matrix $\tilde{\mathbf{M}}^{(p)}$ correspond to the principal directions under a curvature-weighted metric. Specifically, if $\mathbf{u}$ is a left singular vector of $\tilde{\mathbf{M}}^{(p)}$ with singular value σ , it satisfies:

$$\tilde{\mathbf{M}}^{(p)} \tilde{\mathbf{M}}^{(p)\top} \mathbf{u} = \sigma^2 \mathbf{u}. \quad (42)$$

By substituting $\tilde{\mathbf{M}}^{(p)}$ and defining $\mathbf{v} = (\mathbf{G}^{(p)})^{\frac{1}{2}} \mathbf{u}$, we obtain the equivalent generalized eigenproblem:

$$[\mathbf{M}^{(p)} \text{diag}(\alpha) \mathbf{M}^{(p)\top}] \mathbf{v} = \lambda (\mathbf{G}^{(p)})^{-1} \mathbf{v}, \quad \lambda = \sigma^2, \quad (43)$$

which implies that the extracted directions are eigenvectors of the client covariance measured under the curvature-weighted metric $(\mathbf{G}^{(p)})^{-1}$. Equivalently, $\mathbf{v}$ maximizes the Rayleigh quotient [3]:

$$\mathcal{R}(\mathbf{v}) = \frac{\mathbf{v}^\top (\mathbf{M}^{(p)} \text{diag}(\alpha) \mathbf{M}^{(p)\top}) \mathbf{v}}{\mathbf{v}^\top (\mathbf{G}^{(p)})^{-1} \mathbf{v}}. \quad (44)$$

Here, the numerator measures the strength of client updates along $\mathbf{v}$, while the denominator penalizes directions that are not aligned with high-curvature features. This formulation ensures that the dominant spectral components of $\tilde{\mathbf{M}}^{(p)}$ align with geometrically meaningful variations, producing curvature-consistent subspaces for subsequent aggregation.

H Algorithm

The overall algorithm of FedLGS is provided in Alg. 1. Lines 3-12 are the main process for the communication between server and

clients. Lines 17-21 refer to the geometry-aware graph projection module on the client, addressing **CH1**. Lines 7-11 refer to the shared subspace aggregation module on the server, addressing **CH2**.

Algorithm 1 Training procedure of FedLGS

Input: communication rounds T , number of clients K , preserved rank ratio r , client weights $\{\mathbf{w}_k\}_{k=1}^K$

Output: global graph projector θ_g^T

- 1: Server initializes θ_g^0
 - 2: Each client k precomputes $\rho_k(j, i)$ on $\mathcal{G}_k$ by Eq. (35)
 - 3: **for** round $t = 0, 1, \dots, T - 1$ **do**
 - 4: **for** client $k = 1, 2, \dots, K$ **in parallel do**
 - 5: $(\theta_k^{t+1}, \{\gamma_k^{(p)}\}_p) \leftarrow \text{ClientUpdate}(k, \theta_g^t)$
 - 6: **end for**
 - 7: Stack client updates to matrix $\mathbf{M}^{(p)}$ for manifold p
 - 8: Calculate whitened matrix $\tilde{\mathbf{M}}^{(p)}$ by Eq. (19)
 - 9: Extract shared subspace bases $\mathbf{B}_{\text{share}}^{(p)}$ by Eq. (23)
 - 10: Obtain aggregated global graph projector $\theta_g^{(p), t+1}$ by Eq. (28)
 - 11: Combine $\{\theta_g^{(p), t+1}\}$ to the global graph projector θ_g^{t+1} .
 - 12: **end for**
 - 13: **Client executes** (k, θ_g^t) :
 - 14: Receive updated global graph projector $\theta_k^t \leftarrow \theta_g^t$
 - 15: **for** each local epoch $e = 1, 2, \dots, E$ **do**
 - 16: Project onto the product manifold by Eq. (7)
 - 17: Calculate propagation weights $f_{\mathcal{M}_p}$ by Eq. (9)
 - 18: Get the refined node embeddings $\mathbf{z}^{(p)}$ by Eq. (11)
 - 19: Project to the LLM input space by Eq. (12)
 - 20: Optimize the graph projector by Eq. (13)
 - 21: **end for**
 - 22: **return** $(\theta_k^E, \{\gamma_k^{(p)}\}_p)$ to server
-

I Computation and Communication Efficiency

Communication Cost. FedLGS is communication-efficient since each client only uploads projector parameters θ_k and curvature-importance vectors $\{\gamma_k^{(p)}\}_p$, both comparable in size to GNN layers. The LLM is frozen and never transmitted, making each communication round lightweight regardless of the base model size.

Computation Cost. On the client side, geometry-aware refinements on the product manifold cost $O(|\mathcal{E}_k|d_m)$ per layer—comparable to standard GNN message passing. Computing curvature-weighted importance vectors and exponential-map projections adds $O(|\mathcal{E}_k|d_m + |\mathcal{V}_k|d_m)$, which remains linear and lightweight. On the server side, the curvature-based whitening operates entirely on diagonal statistics, allowing efficient element-wise operations with linear cost $O(D_p K)$. The subsequent projector aggregation relies on blockwise SVD. Each whitened projector weight matrix $\Delta \tilde{\mathbf{W}}_{k,\ell}^{(p)} \in \mathbb{R}^{d_\ell^{\text{out}} \times d_\ell^{\text{in}}}$ is decomposed with cost $O(\min\{d_\ell^{\text{out}}, d_\ell^{\text{in}}\}^2 \cdot \max\{d_\ell^{\text{out}}, d_\ell^{\text{in}}\})$, and the second SVD for shared-subspace extraction involves concatenated bases of width Kr_ℓ , costs $O(\min\{d_\ell^{\text{out}}, Kr_\ell\}^2 \cdot \max\{d_\ell^{\text{out}}, Kr_\ell\})$. The following projection and aggregation steps add $O(D_p r_\ell K)$. Since server-side procedures scale linearly with D_p and cubically only with small per-layer dimensions, the overall computation remains well controlled in practice.